Grade 7 Mathematics

Strand 1: Number

Sub-strand 1: Number and Numeration Systems

Lesson 1 — Place Value, Big Numbers, and Rounding

Codes: B7.1.1.1.1 – B7.1.1.1.5, B7.1.1.2

1. What this lesson is about

You already count: 1, 2, 3, … This lesson is about **how big numbers are built**, how to **read and compare** them, and how to **round** them so they are easy to talk about.

Words we will unpack first:

Word	Meaning
Digit	one symbol from 0 to 9
Place value	what a digit is worth because of where it sits
Whole number	0, 1, 2, 3, … (no fractions)
Round	replace a number with a nearby "neater" number
Significant figure	a digit that actually carries meaning
Decimal place	a position after the decimal point (tenths, hundredths, …)

2. Place value — the foundation

When you write **4 725**, each digit's job depends on its position:

```
Thousands | Hundreds | Tens | Units
4 | 7 | 2 | 5
= 4000 | = 700 | = 20 | = 5
```

So $4\ 725 = 4\ 000 + 700 + 20 + 5$.

Going **left** multiplies by 10 each step. Going **right** divides by 10 each step.

```
100 000 | 10 000 | 1 000 | 100 | 10 | 1 | . | 0.1 | 0.01 | 0.001
```

The value of one digit (very common exam question)

In **9 463 285**, what is the **value** of the 6?

Count from the right: 5 (units), 8 (tens), 2 (hundreds), 3 (thousands), **6 (ten-thousands)**.

Value = $6 \times 10\ 000 = 60\ 000$.

*The digit is 6. Its **value** is 60 000. Do not mix these up.*

3. Numbers bigger than one billion

We group big numbers in **threes** from the right:

Group	Name
… billions	1 000 000 000
… millions	1 000 000
… thousands	1 000
… units	1

Example: write 3 050 082 170 in groups: `3` | `050` | `082` | `170` → **three billion, fifty million, eighty-two thousand, one hundred and seventy.**

Example (B7.1.1.1.1 — modelling): model 2 500 000 using blocks/place-value charts: 2 millions + 5 hundred-thousands + 0 + 0 + 0 + 0.

4. Comparing and ordering big numbers

Rule: compare left to right. The first digit where they differ decides.

Example: order these: 4 530 082; 4 350 820; 4 530 802.

1. All start with 4 → look at the next digit: 5, 3, 5. The 3 is smallest → **4 350 820 is smallest**.
2. Now compare 4 530 082 and 4 530 802: same until the last three digits: 082 vs 802.

`082 < 802`.

3. Order: **4 350 820 < 4 530 082 < 4 530 802**.

5. Rounding

5.1 Rounding whole numbers (off, up, down)

- **Round off** to the nearest 10/100/1000… uses the usual "5 or more rounds up" rule.
- **Round up** always goes to the next step up (ceiling).
- **Round down** always goes to the step down (floor).

Example: round 4 738 to the nearest 100.

- Look at the tens digit: 3 (which is 0–4) → round down to **4 700**.
- If it had been 4 758, tens digit 5 → round up to **4 800**.

5.2 Rounding decimals

Round 12.473 to:

- **nearest tenth (1 d.p.)** → look at hundredths (7) → up → **12.5**
- **nearest hundredth (2 d.p.)** → look at thousandths (3) → down → **12.47**
- **nearest thousandth (3 d.p.)** → already exact → **12.473**

5.3 Significant figures (B7.1.1.1.5)

Start counting from the **first non-zero digit**.

Example: 0.006 845

- 1 s.f. → **0.007** (next digit 8 rounds up)
- 2 s.f. → **0.0068**
- 3 s.f. → **0.00685**

6. Twenty fully worked examples

1. Value of 7 in 832 713 → **700**.

2. Value of 6 in 9 463 285 → **60 000**.
3. Write 5 040 007 in words → five million, forty thousand, and seven.
4. Write 3 050 082 in words → three million, fifty thousand, eighty-two.
5. Model 2 500 000 in place-value groups → 2 millions + 5 hundred-thousands.
6. Order 4 530 082; 4 350 820; 4 530 802 → 4 350 820 < 4 530 082 < 4 530 802.
7. Round 4 738 to nearest 100 → **4 700**.
8. Round 4 758 to nearest 100 → **4 800**.
9. Round 12.473 to 1 d.p. → **12.5**.
10. Round 12.473 to 2 d.p. → **12.47**.
11. Round 0.997 84 to 2 d.p. → **1.00**.
12. Round 99.95 to 1 d.p. → **100.0**.
13. Round 84 730 to 2 s.f. → **85 000**.
14. Round 0.006 845 to 3 s.f. → **0.00685**.
15. Round 0.000 090 5 to 2 s.f. → **0.000 091**.
16. Which is larger: 4.7 million or 4 070 000? → 4.7 million (`= 4 700 000`).
17. Write "three million, fifty thousand, eighty-two" in figures → **3 050 082**.
18. A town has 2 400 000 people. Write in words → two million, four hundred thousand.
19. Value of digit 4 in 40 508 216 → **40 000 000**.
20. Round 31 416 to the nearest 1000 → **31 000** (tens digit 1 → down).

7. Eight exercise sets × 5 questions

Spinal note: *questions marked **[Review]** pull from earlier work in this lesson. All others are new practice. After the exercises, full solutions are given.*

Exercise 1 — Place value of digits

1. What is the value of the digit 7 in 832 713?
2. What is the value of the digit 6 in 9 463 285?
3. What is the value of digit 4 in 40 508 216?
4. What is the value of the digit 9 in 7 904 512?
5. **[Review]** What is the value of the digit 5 in 25 006 400?

Exercise 2 — Reading and writing

1. Write 5 040 007 in words.
2. Write 3 050 082 in words.
3. Write "three million, fifty thousand, eighty-two" in figures.
4. Write 3 050 082 170 in words (in groups).
5. **[Review]** Write 38 040 in words.

Exercise 3 — Compare and order

1. Order these: 4 530 082; 4 350 820; 4 530 802.
2. Which is larger: 4.7 million or 4 070 000?
3. Order: 999 999; 1 000 000; 999 099.
4. Order: 2 050 082; 2 050 802; 2 005 820.
5. **[Review]** Order 3 050 082; 3 005 820; 3 500 082.

Exercise 4 — Round whole numbers

1. Round 4 738 to the nearest 100.
2. Round 4 758 to the nearest 100.
3. Round 31 416 to the nearest 1 000.

4. Round 84 730 to the nearest 1 000.
5. **[Review]** Round 198 649 to the nearest 10.

Exercise 5 — Round decimals

1. Round 12.473 to 1 d.p.
2. Round 12.473 to 2 d.p.
3. Round 0.997 84 to 2 d.p.
4. Round 99.95 to 1 d.p.
5. **[Review]** Round 198.635 to 1 d.p.

Exercise 6 — Significant figures

1. Round 0.006 845 to 1 s.f.
2. Round 0.006 845 to 2 s.f.
3. Round 0.006 845 to 3 s.f.
4. Round 84 730 to 2 s.f.
5. **[Review]** Round 0.000 090 5 to 2 s.f.

Exercise 7 — Big-number modelling & value

1. Write 2 500 000 as (millions + hundred-thousands + …).
2. A town has 2 400 000 people. Write the number in words.
3. How many digits are in one billion written out?
4. Write 1 000 000 000 using an index (10 to the power …).
5. **[Review]** Value of the digit 5 in 25 006 400.

Exercise 8 — Mixed / challenge

1. Round 31 416 to the nearest 100.
2. Order 999 999; 1 000 001; 1 000 100.
3. Write 12 345 678 in words.
4. **[Review]** Value of 6 in 9 463 285.
5. **[Review]** Round 12.473 to 3 d.p.

8. Full worked solutions

Solutions — Exercise 1

1. 7 is in the hundreds place → **700**.
2. 6 is in the ten-thousands place → **60 000**.
3. 4 is in the ten-millions place → **40 000 000**.
4. 9 is in the hundred-thousands place → **900 000**.
5. 5 is in the millions place → **5 000 000**.

Solutions — Exercise 2

1. **five million, forty thousand, and seven**.
2. **three million, fifty thousand, eighty-two**.
3. **3 050 082**.
4. **three billion, fifty million, eighty-two thousand, one hundred and seventy**.
5. **thirty-eight thousand and forty**.

Solutions — Exercise 3

1. **4 350 820 < 4 530 082 < 4 530 802**.
2. 4.7 million = 4 700 000 > 4 070 000 → **4.7 million**.
3. **999 099 < 999 999 < 1 000 000**.
4. **2 005 820 < 2 050 082 < 2 050 802**.
5. **3 005 820 < 3 050 082 < 3 500 082**.

Solutions — Exercise 4

1. Tens digit 3 → down → **4 700**.
2. Tens digit 5 → up → **4 800**.
3. Hundreds digit 4 → down → **31 000**.
4. Hundreds digit 7 → up → **85 000**.
5. Units digit 9 → up → **198 650**.

Solutions — Exercise 5

1. Hundredths 7 → up → **12.5**.
2. Thousandths 3 → down → **12.47**.
3. Rounding up carries through → **1.00**.
4. Hundredths 5 → up → **100.0**.
5. Hundredths 3 → down → **198.6**.

Solutions — Exercise 6

1. **0.007**.
2. **0.0068**.
3. **0.00685**.
4. **85 000**.
5. **0.000 091**.

Solutions — Exercise 7

1. `2 000 000 + 500 000`.
2. **two million, four hundred thousand**.
3. 1 000 000 000 has **10 digits**.
4. `10^9`.
5. **5 000 000**.

Solutions — Exercise 8

1. Hundreds digit 4 → down → **31 400**.
2. **999 999 < 1 000 001 < 1 000 100**.
3. **twelve million, three hundred and forty-five thousand, six hundred and seventy-eight**.
4. **60 000**.
5. **12.473**.

9. Cheat sheet

Skill	Rule
Value of a digit	digit × its place
Compare numbers	left to right
Round off	5 or more rounds up
Round up / down	ceiling / floor

Significant figures	start at first non-zero digit

NaCCA CCP B7.1.1 — plain English, British notation.